CS3003: IT Project Management

Surgery No 2 Sample Answers

Q1. Project Scheduling: Network Diagrams and Critical Path Analysis

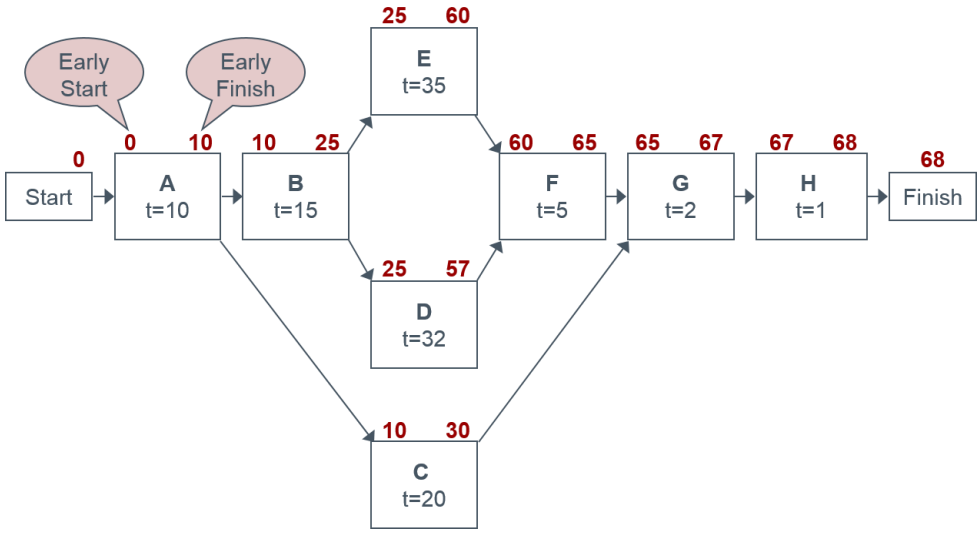
You are given the following activities/tasks and their respective durations and dependencies.

Activity/Task	Duration (Weeks)	Predecessors
A	10	
B	15	A
C	20	A
D	32	B
E	35	B
F	5	D,E
G	2	C,F
H	1	G

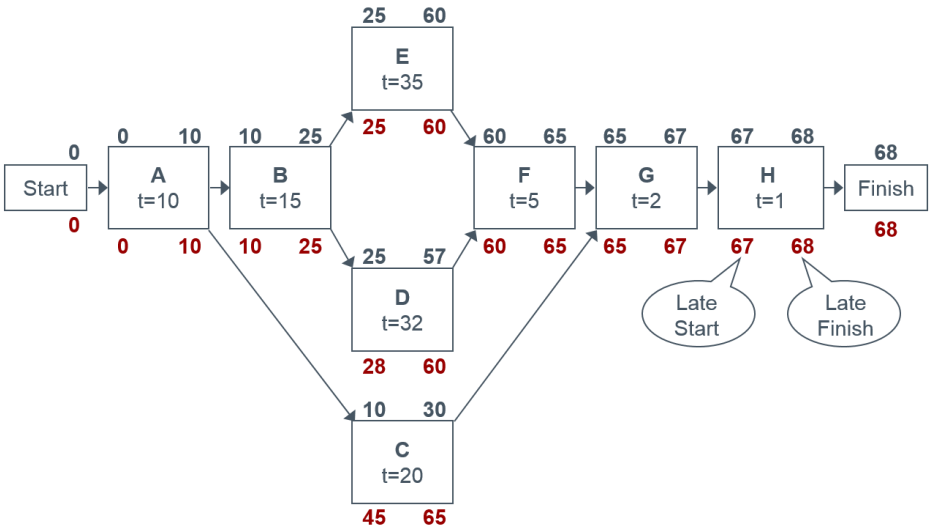
Draw a Network Diagram of the Project, calculate all floats and determine the Critical Path.

Sample Answer

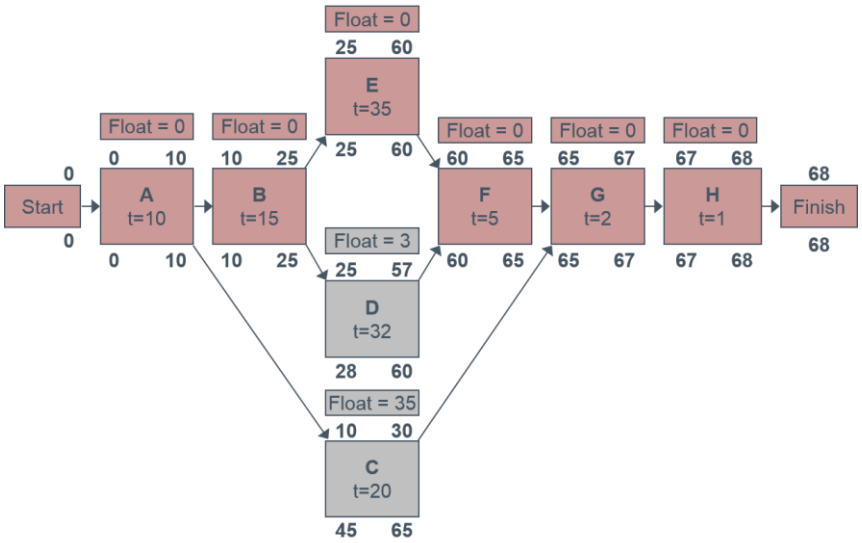
a) Forward Pass



b) Backward Pass



c) Critical Path (A-B-E-F-G-H)



Q2. Project Scheduling: Network Diagrams, Critical Path Analysis, Gantt Chart

Your company has decided to proceed with an IT project and you have been assigned to manage it. You have drawn up an outline of an activity list that includes the following tasks:

A. Build up your team and define a list of requirements to be included in an RFP process for package selection (5 weeks)

B. Assess the alternative packages in the RFP and select the most appropriate. (7 weeks)

C. Specify and acquire the required new hardware and infrastructure for the package. (4 weeks)

D. Test and install all the new hardware and infrastructure. (10 weeks)

E. Modify/Customize and Test the package software. (16 weeks)

F. Install the package software in the production environment (1 week)

G. Specify & obtain the data required to be migrated to the package (4 weeks)

H. Define the key-users training plan. (3 weeks)

I. Train a set of key-users (8 weeks)

J. Define an acceptance test plan (3 weeks)

K. Perform Acceptance Testing (5 weeks)

L. Load data to the new system. (3 weeks)

The dependencies between the 10 tasks listed above are as follows:

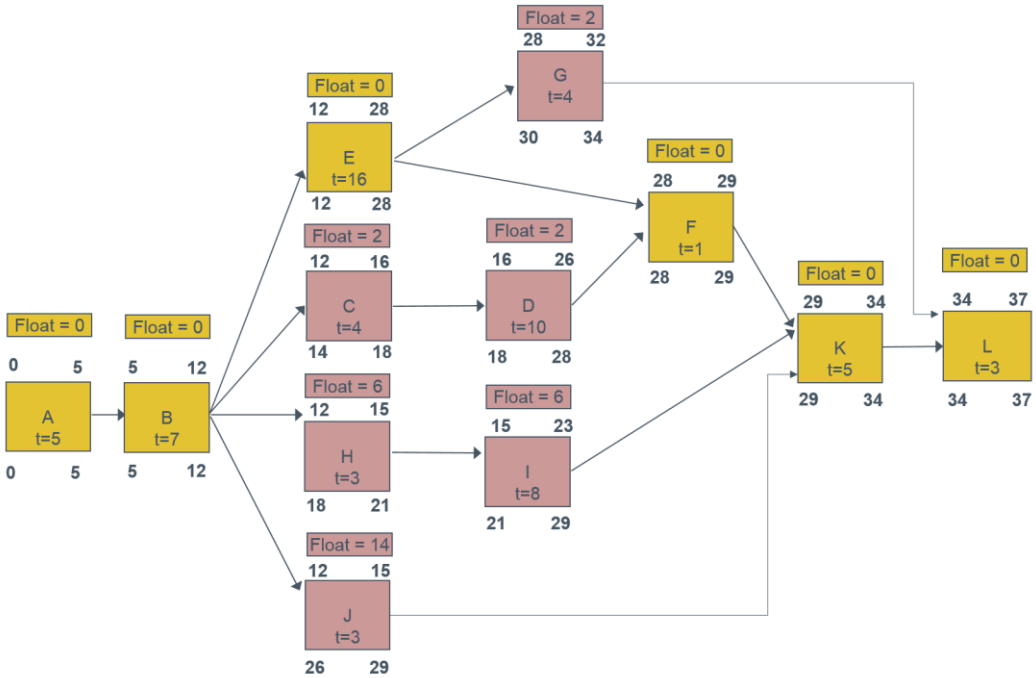
- B depends on A
- C, E, H and J all depend on B
- D depends on C
- F depends on D and E
- G depends on E
- I depends on H
- K depends on F, I and J
- L depends on G and K

a) Draw a Network Diagram of the Project and perform a Critical Path Analysis

b) Draw the full Gantt Chart of the project

Sample Answer

(a) Diagram below. Critical Path: A-B-E-F-K-L



(b) Please see the "Project Libre" tool project file for Gantt Chart.

Q3. Project Variance and Standard Deviation (extra material, not discussed in lectures, not required for the exam)

You are given the following table of a project's activity durations and variances (i.e. optimistic (o), most likely (m) and pessimistic (p) estimates).

Activity/Task	Predecessors	Optimistic	Most Likely	Pessimistic	Estimated Time (ET)	Activity Variance (AV)
A		3	4	11		
B	A	2	5	8		
C	A	3	6	9		
D	B,C	8	12	20		
E	B	3	5	12		
F	D	2	4	7		
G	C	6	9	14		
H	E,F,G	1	2	4		

We have discussed that the estimated time for an activity can be calculated using the PERT formula (also known as beta distribution) as:

Activity **ET (estimated time) = (o + 4m +p) / 6**

Another important formula is called *Activity Variance* and is given by the following formula:

Activity Variance (AV) = $[(p-o)/6]^2$

where p is the most pessimistic time and o the most optimistic time for an activity.

a. Complete the last two columns of the table with the TE and the AV for each of the activities

b. calculate the ***Total Project Variance*** as

   PV = Σ (variances of activities in the critical path)

c. calculate the ***Project Standard Deviation*** as

   Project Standard Deviation (PSV) = sqrt (PV)

Sample Answer

Activity/Task	Predecessors	Optimistic	Most Likely	Pessimistic	Estimated Time (ET)	Activity Variance (AV)
A		3	4	11	5	1.78*
B	A	2	5	8	5	1.00
C	A	3	6	9	6	1.00
D	B,C	8	12	20	12.7	4.00
E	B	3	5	12	5.8	2.25
F	D	2	4	7	4.2	0.69
G	C	6	9	14	9.3	1.78
H	E,F,G	1	2	4	2.2	0.25

*$[(p-o)/6]^2 = [(11-3)/6]^2 = 64/36 = 1.777777777.... = 1.78$

Total Project Variance PV = Σ (variances of activities in the critical path) = Σ (variances of activities A-C-D-F-H) = 1.78+1.00+4.00+0.69+0.25 = 7.72

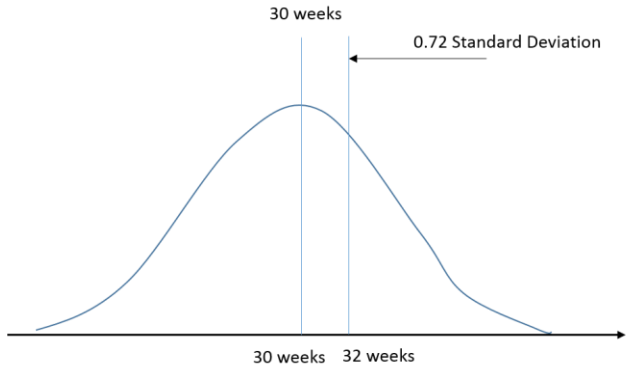
Project Standard Deviation (PSV) = sqrt (PV) = sqrt (7.72) = 2.78 weeks.

We could use the above to perform the following analysis. What is the probability of the project (that has an expected duration of 30 weeks) to finish no later than 32 weeks?

We can use the standard normal equation to determine this probability:

Z= (Target Date – Expected Date of Completion)/PSV = (32 – 30)/2.78 = 0.72

Using a Normal Distribution Table we can conclude that for a Z value of 0.72 the probability is 0.7642. In other words, there is a 76.42% probability that the project will finish on or before 32 weeks.



Finally, one may ask the question, "What is the additional time required so that the project would have a 95% probability of on-time completion?".

This may be calculated as:

Due Date = Expected Date of Completion + (Z x PSV)

In this case for a probability of 95% a Normal Distribution Table indicates that Z = 1.65, thus:

Due Date = 30 weeks + (1.65)x(2.78) = 34.59 weeks.

In other words, if the project could have an additional 4.59 weeks, then there is a probability of 95% that it will be completed on time.

Note: we may also have to consider the impact of activities not in the critical path in such calculations (which has not been the case in the above example)